Lunchbox Allergen Check

Executive Summary: Lunchbox Allergen Check

Problem

Food allergies affect a meaningful number of school-age children in the United States, and parents of these children spend a great deal of time and energy making sure that what their child eats is safe. Every school day involves decisions about lunch: whether to pack food from home, whether the cafeteria menu is safe, and whether a snack that a friend or teacher offers contains something dangerous. Ingredient labels can be long and confusing, and allergens often appear under unfamiliar names. For example, milk may be listed as casein or whey, and wheat may appear in ingredients that parents do not expect. Parents told us that reading labels is tiring and that they worry they will miss something. Many also said that school menus are posted in formats that make it hard to find ingredient information.

We believe this is a real source of anxiety for families and that a simple tool could make daily routines easier.

Solution

Lunchbox Allergen Check is a phone app that lets parents scan a school lunch menu or the label on a packaged snack and quickly see whether it contains their child's allergens. A parent creates a profile for each child and selects the allergens that apply. When the parent points the phone camera at a menu or label, the app reads the text and compares the ingredients against the child's profile. The app then shows a clear result: safe, caution, or avoid. The caution result is used when an ingredient is ambiguous or when a label includes a statement that the product may have been made in a facility that handles an allergen.

The app is designed to be fast and easy to use, even for a parent who is in a hurry in a grocery aisle or at the kitchen table in the morning. We also plan to let parents save results so that they can look back at products they have already checked. In the future, we would like to let parents share their child's profile with relatives, babysitters, and school staff.

Market

Our target customers are parents of school-age children with food allergies. A large number of families in the United States have at least one child with a food allergy, and we believe this represents a substantial market. Because the problem touches almost every school and every community, we think the potential audience is broad. We also see secondary opportunities with schools, summer camps, and after-school programs that want to help families manage allergies more easily.

We have not yet conducted a formal market study. Our current understanding comes from conversations with a small number of parents we know personally and from general reading about food allergies. As part of our next steps, we plan to survey more families and to estimate the number of likely paying users more carefully.

Business Model

We plan to use a freemium model. The free version would let a parent create one child profile and run a limited number of scans each month. A paid subscription would unlock unlimited scans, multiple child profiles, saved history, and sharing features. We are still deciding on the price, but we are considering a monthly fee in the range of a few dollars, with a discount for annual plans.

We also see possible additional revenue from partnerships with food brands that make allergy-friendly products, and from licensing the app to schools. However, we have not discussed these ideas with any potential partners, and we are not yet sure how they would work in practice. For now, we expect subscriptions to be our main source of revenue. We have not built a detailed

model of how many free users would convert to paid users, though we hope that a meaningful share would find the app valuable enough to pay for.

Competition

There are other apps and websites that help people with food allergies, including barcode scanners, ingredient lookup tools, and general nutrition apps. Many of these focus on adults or on general dietary preferences rather than on children and school settings. We believe our app will stand out because it is built specifically for parents of school-age children and because it handles school lunch menus as well as packaged foods.

We recognize that existing apps may add similar features, and that a larger company could copy our approach. Our advantage, we believe, will come from focusing on the needs of families and from making the experience simple and trustworthy. We have not yet tested our app against competing products, and we do not have a patent or other protection for our ideas.

Team

Priya Raghunathan is a junior studying computer science at our university. She will lead the technical development of the app, including the scanning and text recognition features. Marcus Delacroix-Whitfield is a senior studying business. He will lead marketing, partnerships, and planning for the company. Tamsin Okonkwo is a junior studying public health. She will lead our work on food allergy content, including how allergens are named and categorized, and will help us speak with families and health professionals.

Each of us has a personal connection to the problem through friends or family members who manage food allergies. We have formed Lunchbox Allergen Check as a limited liability company. None of us has yet built or launched a consumer product, and we are learning as we go. We plan to seek advice from faculty members and from mentors in the local startup community.

Milestones and Financials

We have not yet reached any formal milestones. We have formed the LLC and developed our concept, and we have begun sketching how the app would look and work. We do not yet have a working prototype, a website, or paying customers.

Over the next twelve months, we hope to build a first version of the app, test it with a group of volunteer families, and use their feedback to improve it. We would then like to release the app to the public and begin to grow our user base. We expect our main costs to be software development, cloud services for text recognition, legal and insurance costs, and marketing. We have not prepared detailed financial projections, but we estimate that we will need several thousand dollars to build a first version and more as we grow. Revenue would begin only after launch, and we cannot yet say when the company might break even.

What We Are Asking For

We are applying to this competition for several reasons. First, we would like funding to build our first prototype and to pay for early costs such as legal fees and software tools. Second, we would value feedback from the judges on our business plan, especially on pricing, competition, and how to reach families. Third, we would welcome introductions to mentors who have experience in consumer apps, food and health, or working with schools. Winning this competition would help us move from an idea to a working product, and we are grateful for the chance to present our plan.